

TWINKIT · REQUIREMENTS

TwinKit requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/twinkit		
Source	docs/03-requirements.md @ 84a120a		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-25	Amish Chadha	Initial scaffold

TwinKit requirements

These are first-pass requirements for the concept. Targets are proposals for review. Status is judged from the first-order estimates in TWK-PRC-001 and will be checked by calculation at TRL 3. One requirement (R14, cost) is not met, and two (R12, thermal; R13, setup time) are not yet shown.

Table 1. Requirements.

ID	Requirement	Target	Status at TRL 2	Verification (TRL 3 or later)
R1	Receive readings from lab sensor nodes	50 nodes at 1 reading per 5 min each, under 1 % of uplinks lost at SF7 to SF9	Met by estimate	Airtime and collision calculation
R2	Accept common inputs	LoRaWAN (EU868 or US915), MQTT over Ethernet or Wi-Fi, HTTP POST	Met by design	Design review
R3	Store history on the gateway	1 year for 50 nodes x 10 fields at 5 min intervals	Met by estimate (about 1.6 GB of 64 GB)	Storage calculation
R4	Link data to the model	Every sensor channel mapped to a named part (BOM number) of the project's build123d model and to an expected value or calculation, in a validated schema	Met by design	Schema review on the example twin
R5	Compare measured and expected	Residual and pass or flag state shown within 60 s of a reading arriving; flag when outside a stated tolerance for a stated time	Met by estimate (under 5 s)	Latency estimate; later bench run
R6	Show the twin in 3D	The project's glTF model, colored by live values, in a current browser on a mid-range laptop or phone	Met by design	Design review
R7	Work offline	All functions run with no internet; optional upstream sync buffers at least 7 days	Met by design	Design review
R8	Low-voltage power	9 to 30 V DC input, 12 V nominal; average draw 8 W or less	Met by estimate (about 6 W)	Power budget
R9	Ride through outages	At least 30 min on backup and a clean shutdown before the pack is empty	Met by estimate (about 2 h)	Backup energy calculation
R10	Compact mounting	Fits a TS35 DIN rail; gateway, converter and backup within 20 modules (350 mm) of rail	Met (about 17 modules in the massing model)	Massing model
R11	Secure by default	Unique credentials set at first boot, TLS on the dashboard, no ports open beyond the local network by default	Met by design	Design review; later configuration audit
R12	Operate in a lab or equipment room	Ambient 0 to 40 °C without CPU throttling	Not yet shown (enclosure air about 20 K above ambient, estimate)	Thermal calculation
R13	Quick setup	From a flashed card to the first live reading on the dashboard in 60 min or less, following the guide	Not yet shown	Timed trial at TRL 4
R14	Low cost and buildable	Parts \$200 or less; no custom PCB	Not met (about \$285; no custom PCB is met)	Priced BOM
R15	Open and exportable	All software open source; data exportable as CSV and through an open API	Met by design	Design review

ID	Requirement	Target	Status at TRL 2	Verification (TRL 3 or later)
R16	Monitoring only	No control outputs; the gateway cannot actuate the monitored system	Met by design	Design review

■ Assumptions

- A sensor reading is a LoRaWAN uplink of about 20 bytes of payload. Airtime is about 60 ms at SF7 and about 1.3 s at SF12, both at 125 kHz bandwidth.
- Uplinks spread evenly over 8 channels and arrive at random (pure ALOHA).
- A stored value takes about 30 bytes including index overhead, before any database compression.
- The gateway draws about 6 W: single-board computer about 4 W under light load, concentrator about 1 W, converter and UPS losses about 1 W.
- The backup pack delivers 80 % of its 19.2 Wh rating.